Does Height Matter for scoring in the NBA?

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Abstract

Does height matter for scoring in the NBA? This project examines whether taller players score more points per game (PPG), using player-level statistics from the NBA website, including height, PPG, and minutes played. Position and minutes played will be added as controls in later stages. So far, we have created preliminary visualizations comparing height and PPG across seasons, and we plan to adjust based on other external factors. We have found that the overall correlation between height and PPG is fairly weak, especially when considering all NBA players. The correlation is slightly stronger for the elite scorers, probably because of a few outliers that heavily skew the data. As we look into more variables we believe that the height variable will weaken and other variables will emerge as better explanations as to why a player might score PPG.

Introduction

Basketball is often one of the major sports thought to be heavily reliant on tall players. Height helps with rebounding, shot selection, playing defense, and these have led to the common perception that short players are at an inherent disadvantage in the NBA. But in the NBA, there are several prolific scorers that aren't the tallest, like Steph Curry. This raises our central question, does height actually matter for scoring in the NBA?

The central question of this project is whether a player's height is associated with the number of points he scores per game (PPG). Using individual statistics from the NBA website, we compare height and PPG across multiple seasons. The preliminary visualizations so far suggest a weak relationship across all players, with a possible stronger pattern among elite scorers, defined as players averaging 25 or more points per game. Because few players reach this threshold, a handful of elite tall players may be driving this pattern, and we want to eventually examine all the factors that influence this.

Our goal is to test a hypothesis. We will formally test whether the relationship between height and PPG is statistically significant, and we will measure how large the relationship is. The null hypothesis is that height has no linear relationship with PPG, and the alternative is that a relationship exists. In later stages, we plan to control for other factors that affect scoring, such as minutes played and position, to see whether height still matters once these are accounted for.

This analysis may benefit several groups. College coaches and scouts can use it to judge how much weight height deserves when evaluating scorers. Eventual predictive models can assess how much height of players will affect their scoring performances in the major leagues. A relationship might help fans and fantasy basketball players use it to understand which players tend to score more. Aspiring basketball players, especially those who won't reach the heights of the average NBA player, can see evaluate whether height is a barrier to becoming a high scorer and adapt their game based on this analysis.

Executive Summary

- Based on our visualizations of the entire NBA, there is a weak relationship between height and PPG, showing that overall height does not affect scoring performance.

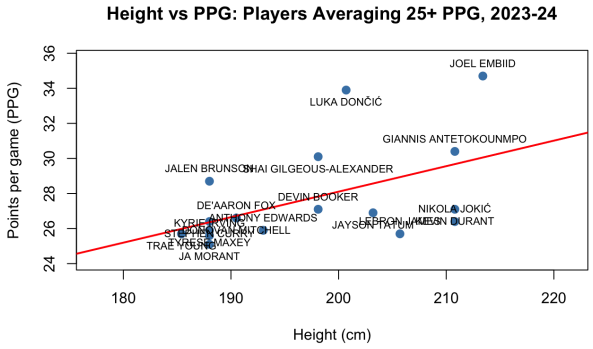
- Based on our visualizatinos of the elite scorers, there is a possibility that there is a weak relationship between height and ppg. Overall, when looking at the elite scorers only, the line of best fit slightly trends up.
- Across the 3 seasons, there are generally less than 20 elite scorers per season, showing the low sample size with outliers affecting the data heavily. In the future, we should look at how different charateristics/variables affect the data.

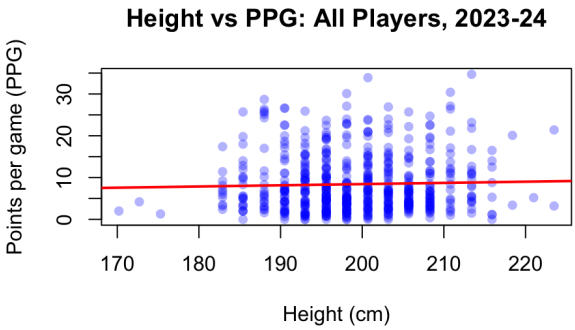
Data

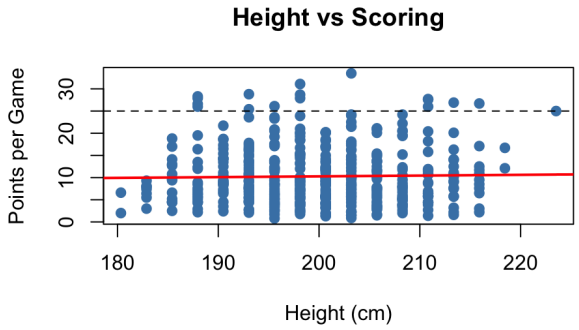
The data for this project come from player statistics published on the NBA website (NBA.com), which we downloaded as CSV files. For each season we use one file of player biographical information and one file of player statistics. We analyze three seasons: 2023-24, 2024-25, and 2025-26.

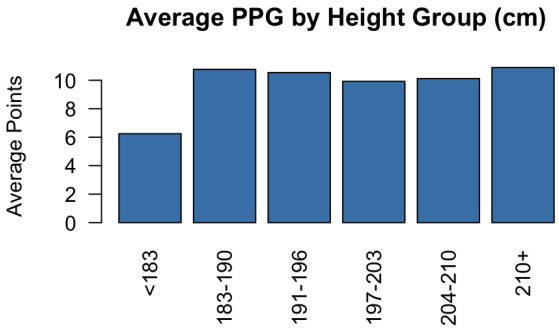
The variables available are general stats like name, height, position, ppg, minutes played, and currently the variables we filtered are generally the important ones that we are looking at. Each year has a different number of observations (over 500) which accounts for all the NBA players that year. The steps performed to clean the data were generally to clean players that did not score over 25 ppg, and others that did not average many minutes or make NBA rosters.

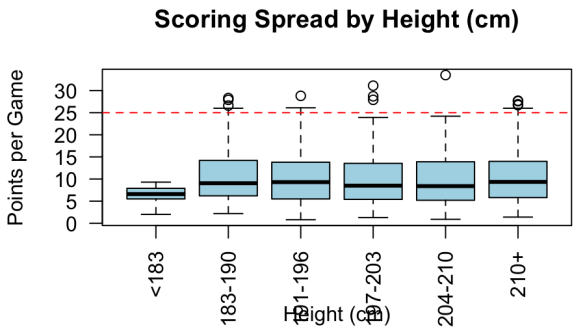
Visualization

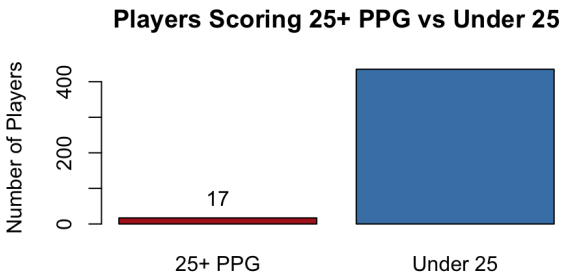


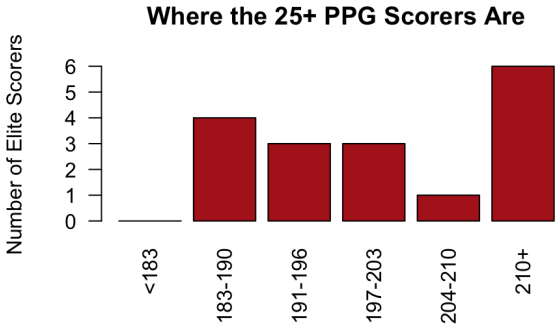












Our preliminary visualizations can be found in the Stat107_Project.RMD. The graphs that apply to the entire NBA suggest a very weak/no correlation between height and PPG. This challenges our hypothesis that there is a relationship between height and scoring in the NBA. The graphs that filter for elite scorers only provide a weak/medium correlation between height and scoring, although these do not necessarily support our hypothesis due to the low sample size and tendency for outliers to skew the data. We aim to generate further data visualizations that illustrate the effect of minutes played, and position (aswell as other unforseen factors) on the ability for a player to score. Eventually, we will probably find that other factors are more indicative of a player's PPG than height.

Analysis

Our analysis will test whether height is associated with scoring and how large that association is, using correlation tests and then with other models to account for our unused factors. We will run the visualiations on each season, then compare amongst them to see if the conclusions are consistent.

1. Correlation test will compare the relationship between height and PPG, with the null hypotehsis being that there is no linear relationship, and the alternative that a relationship exists. We will report the correlation co-efficient, confidence interval, p-value, and run it separately based on all NBA players versus the elite scorers.

2. We will also run a linear regression to create an estimation of how many additional points per game are associated with each added centimeter of height. We will use the R-squared coefficient to determine how much variatino in PPG that can be explained by height. We might also plan on doing more regressions with controls, to check whether the "height effect" remains once minutes played and positions are accounted for.

3. Because our boxplots show outliers in taller groups, we will also compare the effects of how a small number of superstar players might influence the results, and temper them amongst our results. In

addition, we have to take into account form dips based on seasons, as there might potentially be different effects of heights during each season. We expect to find a weak relationship between height and scoring once all our variables are accounted for, but we also must acknowledge the weaknesses to our study. Primarily, the elite group of players is small and is heavily influenced by 1 or 2 outliers, and because this is observational data, a significant result would end up showing an association, rather than causation.

Summary of Individual Contribution

Stefan Assigned to the 25/26 NBA season. Analyzed the csv data in the data cleaning file, and produced 5 visualizations to represent the relationship between PPG and height in two populations, the entire NBA and the elite scorers. Summarized his findings, and helped to author the Final Report.

Damia Assigned to the 24/25 NBA season. Analyzed the csv data in the data cleaning file, and produced _____ visualizations to represent the relationship between PPG and height in ___ populations. She then documented her maintakeaways from her visualizations and how they support our main hypothesis.

#Nolan Assigned to the 23/24 NBA season. Created the data cleaning functions, and processsed the data before creating 2 visualizations that summarized the difference between the NBA overall and elite scorer populations. Completed the README file and the FinalReport rmd.